

Sustainable ROV-Based Sonar System

for Deep Scattering Layer (DSL) Monitoring
IEEE Hackathon 2026 | Concept Document

71 Project Summary

This project proposes a sustainable, solar-powered underwater Remotely Operated Vehicle (ROV) system for long-term monitoring of the Deep Scattering Layer (DSL) — a dense biological stratum found at 200–1000 m depth in the open ocean. The ROV is tethered via cable to a surface-floating buoy equipped with solar panels, enabling continuous green-energy operation over weeks to months. A multi-frequency active sonar system (38 / 120 / 200 kHz) provides non-invasive biomass estimation of zooplankton, krill, and pelagic fish, while a rolling z-score anomaly detection algorithm flags potential illegal fishing events. The design minimises acoustic disturbance to marine mammals and eliminates the need for destructive trawl surveys.

2. Problem Statement

Current ocean monitoring methods suffer from three critical limitations:

- **Acoustic pollution:** Conventional fisheries sonar operates at 200–210 dB re 1 μ Pa — well above NOAA Level A/B marine mammal harassment thresholds.
- **Destructive assessment:** Trawl surveys cause by-catch and benthic habitat damage, providing only snapshots rather than continuous data.
- **Range limitation:** Echo-friendly passive sonar cannot quantify DSL biomass density or vertical extent without a powered tethered platform.
- **Mission duration:** Untethered AUVs are battery-limited to hours; long-term continuous monitoring is not feasible without an external energy source.

3. Proposed Solution

- **Solar Floating Buoy (Aqua Guardian):** Conical buoy with tilted solar-panel array provides continuous renewable power via tether cable to the submerged ROV — enabling indefinite mission duration without vessel support.
- **Open-Frame ROV:** Aluminium lattice frame with 4 horizontal + 2 vertical T200 thrusters, integrated sonar dome (sphere transducer), operating at 200–1000 m depth.
- **Eco-Friendly Multi-Frequency Sonar:** 38 kHz (fish with swim bladders) / 120 kHz (krill) / 200 kHz (zooplankton); source levels 25–30 dB below conventional; 1% adaptive duty cycle — 30x reduction in cumulative acoustic exposure.
- **Illegal Fishing Detection:** Rolling z-score anomaly detector on 30-day DSL biomass time series; alert triggered at $z < -2$ (>50% biomass drop within 2–4 days).

4. System Architecture

Solar Buoy	Aqua Guardian cone → solar panels → DC converter → tether cable
Tether	Power + bidirectional data link (ROV ↔ surface / satellite uplink)
ROV Frame	Open Al frame, 4x horiz. + 2x vert. BlueRobotics T200 thrusters
Sonar Array	38 / 120 / 200 kHz transducers + sonar dome; 7° narrow beamwidth
Signal Processor	Matched-filter → Sv computation → freq. differencing → DSL tracking
Anomaly Engine	5-day rolling baseline; z-score alert at $z < -2$ per frequency band
Data Uplink	Tether → buoy → satellite/RF link to shore station or research vessel

The buoy–ROV tether configuration directly resolves the range limitation of standalone eco-friendly sonar while maintaining a zero-emission energy profile.

5. Implementation Method

Sonar Design Parameters

Frequencies	38 kHz (fish) 120 kHz (krill) 200 kHz (zooplankton)
Source Level	175–180 dB re 1 µPa @ 1 m (vs 200–210 dB conventional)
Pulse Width	1 ms eco mode / 10 ms conventional mode
Duty Cycle	1% eco vs 30% conventional → 30x reduction in acoustic exposure
Beamwidth	7° narrow beam — spatially limits ensonified volume
Calibration	Standard sphere calibration per ICES CRR No. 326

Signal Processing Pipeline

Step	Operation
1. Raw Acquisition	Digitised echo returns at 38 / 120 / 200 kHz simultaneously.
2. Noise Filtering	Matched-filter pulse compression + ambient noise subtraction.
3. Volume Backscatter	S_v (dB re m^{-1}) computed per depth bin via sonar equation.
4. Freq. Differencing	$\Delta S_v = S_v(120) - S_v(38)$: positive → krill dominant; negative → fish dominant.
5. DSL Detection	Gaussian peak fitting to S_v depth profile; automated layer tracking.
6. Biomass & Carbon	$S_v \rightarrow$ cross-section $\rightarrow$ density (kg/m^2); carbon flux via $RQ = 0.85$.

6. Results and Comparison

Parameter	Conventional Sonar	Proposed Eco-Sonar
Source Level	200–210 dB re 1 µPa	175–180 dB re 1 µPa
Duty Cycle	~30%	1% adaptive
Acoustic Energy	Baseline (100%)	~0.1% (1000x less)
Mission Duration	Hours (battery AUV)	Weeks–Months (solar)
Biomass Method	Trawl — destructive	Non-invasive acoustics
Marine Mammal Risk	High — above NOAA limits	Low–Managed (all bands)
Carbon Footprint	Diesel vessel required	Zero-emission (solar)
Fishing Alert	None	z-score anomaly detector

7. Impact

- **Climate Science:** Continuous carbon flux measurement from diel vertical migration quantifies the biological pump — a critical and poorly constrained component of the ocean carbon cycle.
- **Fisheries Management:** Non-invasive biomass mapping replaces destructive trawl surveys, enabling sustainable stock assessment for krill, zooplankton, and pelagic fish.
- **Illegal Fishing Enforcement:** Near-real-time biomass anomaly alerts provide actionable intelligence for maritime authorities monitoring MPAs and restricted fishing zones.
- **Marine Conservation:** All operating frequencies kept below NOAA Level A/B harassment thresholds, protecting cetaceans and pinnipeds across all frequency bands.

8. Limitations

- **Weather Dependency:** Solar buoy output is reduced during prolonged cloud cover or storms; backup battery buffer not yet sized.
- **Tether Depth Limit:** Cable drag and weight constrain maximum depth; 1000 m target requires high-strength low-drag tether not yet prototyped.
- **Target Strength Uncertainty:** S_v -to-biomass conversion relies on empirical models carrying $\pm 3\text{--}5$ dB uncertainty across species and size distributions.
- **False Positives:** Natural events (upwelling, storms) can produce rapid biomass redistribution resembling fishing depletion signatures.
- **Prototype Status:** ROV and buoy are at initial CAD design stage; hydrodynamic testing and field validation have not yet been conducted.

9. Repository and Reproducibility

How to run them : open simulation and copy the code on Matlab

References

- [1] NOAA (2018). *Technical Guidance for Assessing the Effects of Anthropogenic Sound on Marine Mammal Hearing*. NOAA Technical Memorandum NMFS-OPR-55.
- [2] Foote, K. G., et al. (2007). Protocols for calibrating multibeam sonar. *J. Acoust. Soc. Am.*, 117(4), 2013–2027.
- [3] ICES (2016). *Calibration of Acoustic Instruments*. ICES Cooperative Research Report No. 326.
- [4] Simmonds, J., & MacLennan, D. (2005). *Fisheries Acoustics: Theory and Practice* (2nd ed.). Blackwell Science, Oxford.
- [5] Stanton, T. K., & Chu, D. (2000). Review and recommendations for modelling acoustic scattering by fluid-like elongated zooplankton. *ICES J. Mar. Sci.*, 57(4), 793–807.
- [6] Strömberg, K. H. P., et al. (2009). Relationship between acoustic backscattering target strength and carbon content of mesopelagic organisms. *ICES J. Mar. Sci.*, 66(6), 1202–1211.
- [7] Childress, J. J., & Thuesen, E. V. (1992). Metabolic potential of deep-sea animals. *Deep-Sea Research*, 39(1), 205–213.
- [8] Proud, R., et al. (2017). From siphonophores to deep-sea fish: quantifying biodiversity below the ocean surface using acoustics. *Progress in Oceanography*, 158, 220–230.
- [9] Kaartvedt, S., et al. (2012). Use of bottom-mounted echo sounders in observing subsurface fish behaviour. *Mar. Ecol. Prog. Ser.*, 449, 263–273.
- [10] CCAMLR (2010). *Krill Acoustic Survey Methods*. SC-CAMLR-XXIX, Commission for the Conservation of Antarctic Marine Living Resources.
- [11] Blue Robotics (2024). *T200 Thruster Technical Documentation*. Blue Robotics Inc. <https://bluerobotics.com>
- [12] Mauchline, J. (1980). *The Biology of Mysids and Euphausiids*. Advances in Marine Biology, Vol. 18. Academic Press, London.